ExtremaFuse for Adaptive Compression of Frame-Level Speech Representations

Anonymous ACL submission

Abstract

We study whether content-adaptive frame fusion can shorten speech representations while preserving phoneme recognition quality. ExtremaFuse selects centers at local extrema of the cosine-distance curve between adjacent frames and learns to combine a five-frame neighborhood around each center. We evaluate the method using an offline Conformer transducer trained on 460 hours of LibriSpeech.

1 Introduction

A fixed stride reduces the time resolution of speech at regular intervals. Acoustic transitions and stable intervals need not occupy equal spans of a recording. We ask whether choosing fusion centers from the signal can reduce decoding cost at comparable phone recognition quality, and how the resulting error profile varies with speaking rate.

2 Related Work

Time reduction for speech models. Pooling and adjacent-token merging provide relevant comparisons for transducers (Wang et al., 2024; Li et al., 2023). Work on speech language models and Whisper also examines compression at multiple positions (Bhati et al., 2025; Xiang et al., 2026; Cho et al., 2026).

Local acoustic change and phone evaluation. Phonetic segmentation studies use changes in speech to locate candidate boundaries (Kreuk et al., 2020; Yantosca and Cheng, 2025). Phone recognition studies also motivate analyzing which distinctions a system preserves beyond its aggregate error rate (Bharadwaj et al., 2026).

3 ExtremaFuse

3.1 Sequence interface

The input is an ordered sequence $H = (h_1, \ldots, h_T)$ of valid D-dimensional frames. The output has the same feature dimension and a content-dependent length $K \leq T$. Downstream layers consume the new length and mask. Figure 1 shows how the original frames enter both selection and fusion.

Input	$H = (h_1, \ldots, h_T)$
Select	$H \longmapsto (d_1, \ldots, d_{T-1}) \longmapsto C$
Fuse	$(H, C) \longmapsto (z_{c_1}, \ldots, z_{c_K})$

Figure 1: ExtremaFuse at one insertion site. Distances determine an ordered set C of centers. Each z_c pools the original features around that center with learned weights. The next stage receives K frames of dimension D.

3.2 Selecting fusion centers

For adjacent nonzero frames, define $d_t = 1 - \cos(h_t, h_{t+1})$. A local maximum at p selects frame $p + 1$ and a local minimum at v selects frame v. We merge duplicate positions and retain chronological order. A nonempty sequence with no detected extremum uses its first frame as the center.

3.3 Learning local fusion weights

For a center c, a network operating on its five-frame neighborhood predicts weights over the valid positions W_c. Each output is a convex combination,

$$z_c = \sum_{j \in W_c} \alpha_{cj} h_j, \quad \alpha_{cj} \geq 0, \quad \sum_{j \in W_c} \alpha_{cj} = 1. \tag{1}$$

Windows may overlap. Separate instances can operate on different feature dimensions and learn their weights from the downstream objective.

4 Experimental Setup

4.1 Task and systems

We use train-clean-100 and train-clean-360 from LibriSpeech (Panayotov et al., 2015) for English phoneme recognition with 69 stress-marked ARPABET phones. The evaluation system is an offline

System	Input projection	Encoder width	Params (M)	test-clean PER (%)	test-clean CPU RTF	test-other PER (%)
Fixed stride 4 (E09)	128	512	29.58	5.7427	0.03971	–
Fixed stride 4 (E01)	32	128	4.21	6.1289	0.03116	–
Layer 5 fusion (E06)	128	128	4.24	4.7809	0.05062	–
Mel + layer 5 fusion (E10)	128	128	4.25	5.6882	0.02450	–
Uniform pooling at both sites	128	128	–	–	–	–

Table 1: System configurations and measured cost under one shared CPU protocol. E09 matches the per-frame projection and E01 matches the backbone width. The uniform pooling control and test-other evaluations are pending.

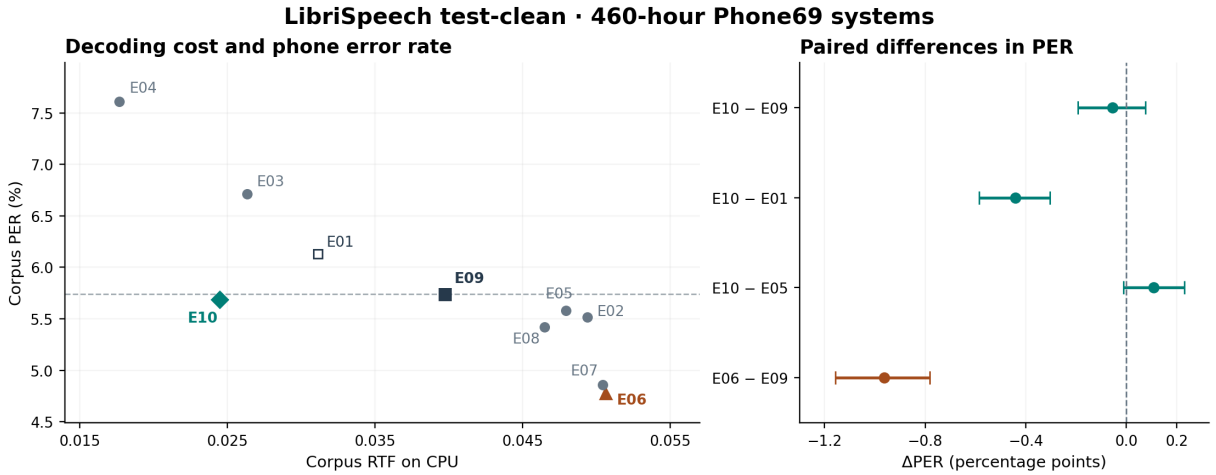


Figure 2: Observed test-clean configurations (left) and paired PER differences (right). The horizontal line marks E09 PER. The intervals resample 40 speakers jointly across systems, with 10,000 draws. RTF comes from one complete timing pass per system. The configurations are listed in Appendix A.

Conformer encoder (Gulati et al., 2020) with a transducer objective (Graves, 2012). In the two-stage system, one fusion module precedes the Mel input projection and the other follows encoder layer 5. E09 matches E10's 128-dimensional per-frame input projection. Its four-frame stack gives a 512-dimensional backbone. E01 matches E10's 128-dimensional backbone with a 32-dimensional input projection. Table 1 gives their parameter counts.

4.2 Uniform pooling control

The trainable system surrounding the control has the same two insertion sites and backbone width as the two-stage ExtremaFuse system. The proposed control partitions each valid sequence into contiguous, disjoint segments and averages the vectors in each segment. Its two retention ratios are to be calibrated on dev-clean and then frozen.

4.3 Quality, cost, and uncertainty

Corpus phone error rate (PER) is the total edit count divided by the reference phone count. Corpus real-time factor (RTF) is total model decoding wall time divided by total audio duration. The current CPU comparison uses four compute threads, one inter-op thread, FP32, batch size one and beam width 20. Timing includes the encoder, predictor, joiner and beam search.

5 Results

5.1 Quality and decoding cost

Table 1 compares observed operating points. Relative to the projection-matched baseline, the two-stage system used 38.32% less decoding time on test-clean. The after-layer-5 system had the lowest PER in this set and used 27.47% more decoding time than the same baseline.

The test-clean PER difference between E10 and E09 is -0.054 percentage points, with a paired 95% speaker-bootstrap interval of $[-0.192, +0.076]$. The width-matched comparison with E01 is an additional check on model capacity. Figure 2 gives the measured configurations.

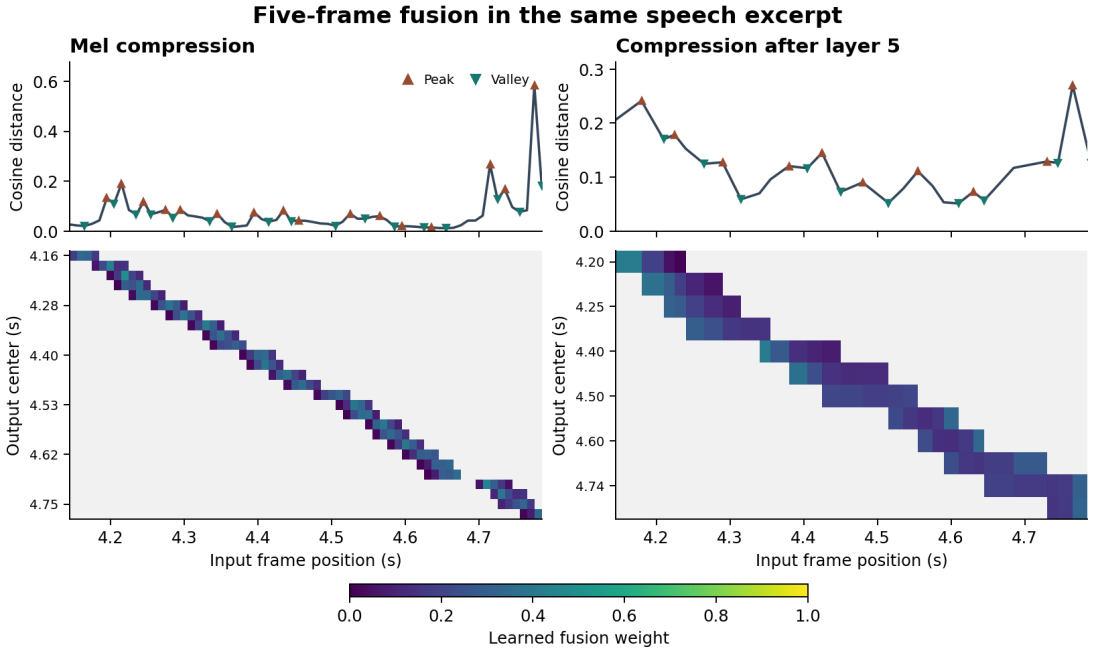


Figure 3: Measured distance curves and fusion weights for the same excerpt at the Mel input and after layer 5. Each row in the lower panels represents an output center. Layer-5 positions are expressed on the Mel time axis; their features already incorporate encoder context.

5.2 Independent evaluation and pooling

The independent comparison will evaluate the frozen systems under a quality margin specified before examining test-other results.

6 Analysis

6.1 The cost of the second fusion stage

E05 provides the Mel-only comparison for E10. Both systems produced 764,950 Mel-stage outputs across test-clean. After layer 5, E10 retained 307,404 steps, a further reduction of 59.81%. PER changed from 5.5782% to 5.6882%. The systems were trained separately.

6.2 Window coverage and a matched-length intervention

Equation 1 gives a bound on local distortion,

$$\|z_c - h_c\|_2 \leq \sum_{j \in W_c} \alpha_{cj} \|h_j - h_c\|_2. \qquad (2)$$

The bound concerns the representation at one center. Recognition quality is measured separately with PER. Across 100 test-clean utterances sampled from ten speech-rate bins, the layer-5 fusion windows covered 98.24% of that stage's input positions. At least one weight of 0.05 or more was assigned to 97.03% of the positions. Figure 3 shows the weights for one excerpt. Keeping E10 frozen and preserving each utterance's layer-5 output length, we replaced that stage with uniform pooling. On the same sample, the reference negative log-likelihood changed from 0.1423 to 0.1435 nats per phone. The paired 95% interval for this change was $[-0.00241, +0.00499]$.

6.3 Speaking rate and the error profile

For each utterance, speaking rate is the number of reference phones per minute of audio, including pauses. Relative to E09, E10 made 13 more errors in the slow group, 48 more in the middle group and 164 fewer in the fast group on test-clean. In the fast group it made 504 fewer deletions, 85 more substitutions and 255 more insertions. The fast-minus-slow difference in the PER change was -0.273 percentage points, with a paired 95% interval of $[-0.697, +0.154]$.

7 Conclusion

The current test-clean comparison identifies a two-stage operating point with lower measured CPU cost and a similar observed PER to the projection-matched baseline.

Limitations

The current evidence concerns a single training seed, one offline recognition backbone, English

read speech and a specific CPU decoding protocol. The historical checkpoints differ in optimization histories. Further experiments would be needed to establish effectiveness across other backbones and tasks.

Ethical Considerations

The evaluation uses English audiobook recordings and stressed phone labels. Application to other speaking styles or accents would require corresponding evaluation data.

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A Configurations and Reproducibility

The comparison includes two fixed-stride baselines (E01 and E09), layer-2 fusion with strides 1, 2 and 4 (E02–E04), Mel-only fusion (E05), layer-5 and layer-7 fusion (E06 and E07), layer-2 neighborhood retention followed by layer-5 fusion (E08), and the two-stage configuration (E10).

B Additional Diagnostics

This appendix is reserved for the full rate-bin counts, phone-level error breakdowns and the selection rule for the illustrated excerpt.